

Kapitel 1

Six proofs of the infinity of primes

Theorem 1.1 (Euclid's proof). *A finite set $\{p_1, \dots, p_r\}$ cannot be the collection of all prime numbers.*

Lean: [infinity_of_primes](#)

Beweis. For any finite set $\{p_1, \dots, p_r\}$ of primes, consider the number $n = p_1 p_2 \cdots p_r + 1$. This n has a prime divisor p . But p is not one of the p_i 's: otherwise p would be a divisor of n and of the product $p_1 p_2 \cdots p_r$, and thus also of the difference $n - p_1 p_2 \cdots p_r = 1$, which is impossible. So a finite set $\{p_1, \dots, p_r\}$ cannot be the collection of all prime numbers. $\square$

Theorem 1.2 (Second Proof). *Any two Fermat numbers $F_n := 2^{2^n} + 1$ are relatively prime.*

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Beweis. Let us first look at the Fermat numbers $F_n = 2^{2^n} + 1$ for $n = 0, 1, 2, \dots$. We will show that any two Fermat numbers are relatively prime; hence there must be infinitely many primes. To this end, we verify the recursion

$$\prod_{k=0}^{n-1} F_k = F_n - 2,$$

from which our assertion follows immediately. Indeed, if m is a divisor of, say, F_k and F_n (with $k < n$), then m divides 2, and hence $m = 1$ or 2. But $m = 2$ is impossible since all Fermat numbers are odd. To prove the recursion we use induction on n . For $n = 1$, we have $F_0 = 3$ and $F_1 - 2 = 3$. With induction we now conclude

$$\prod_{k=0}^n F_k = \left(\prod_{k=0}^{n-1} F_k \right) F_n = (F_n - 2) F_n = (2^{2^n} - 1)(2^{2^n} + 1) = 2^{2^{n+1}} - 1 = F_{n+1} - 2.$$

$\square$

Theorem 1.3 (Third Proof). *There is no largest prime.*

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Beweis. Suppose $\mathbb{P}$ is finite and p is the largest prime. We consider the so-called *Mersenne number* $2^p - 1$ and show that any prime factor q of $2^p - 1$ is bigger than p , which will yield the desired conclusion. Let q be a prime dividing $2^p - 1$, so we have $2^p \equiv 1 \pmod{q}$. Since p is prime, this means that the element 2 has order p in the multiplicative group $\mathbb{Z}_q \setminus \{0\}$ of the field $\mathbb{Z}_q$. This group has $q - 1$ elements. By Lagrange's theorem, we know that the order of every element divides the size of the group, that is, we have $p \mid q - 1$, and hence $p < q$. $\square$

Theorem 1.4 (Fourth Proof). *The prime counting function is unbounded*

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Beweis. Let $\pi(x) := \#\{p \leq x : p \in \mathbb{P}\}$ be the number of primes that are less than or equal to the real number x . We number the primes $\mathbb{P} = \{p_1, p_2, p_3, \dots\}$ in increasing order. Consider the natural logarithm $\log x$, defined as

$$\log x = \int_1^x \frac{1}{t} dt.$$

Now we compare the area below the graph of $f(t) = \frac{1}{t}$ with an upper step function. (See also the appendix for this method.) Thus for $n \leq x < n+1$ we have

$$\log x \leq 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1} + \frac{1}{n} \leq \sum \frac{1}{m},$$

where the sum extends over all $m \in \mathbb{N}$ which have only prime divisors $p \leq x$.

Since every such m can be written in a unique way as a product of the form $\prod_{p \leq x} p^{k_p}$, we see that the last sum is equal to

$$\prod_{p \in \mathbb{P}, p \leq x} \left(\sum_{k \geq 0} \frac{1}{p^k} \right).$$

The inner sum is a geometric series with ratio $\frac{1}{p}$, hence

$$\log x \leq \prod_{p \leq x} \frac{1}{1 - \frac{1}{p}} = \prod_{p \leq x} \frac{p}{p-1} = \prod_{k=1}^{\pi(x)} \frac{p_k}{p_k-1}.$$

Now clearly $p_k \geq k+1$, and thus

$$\frac{p_k}{p_k-1} = 1 + \frac{1}{p_k-1} \leq 1 + \frac{1}{k} = \frac{k+1}{k},$$

and therefore

$$\log x \leq \prod_{k=1}^{\pi(x)} \frac{k+1}{k} = \pi(x) + 1.$$

Everybody knows that $\log x$ is not bounded, so we conclude that $\pi(x)$ is unbounded as well, and so there are infinitely many primes. $\square$

Theorem 1.5 (Fifth Proof). *The set of primes $\mathbb{P}$ is infinite.*

Lean: [infini-ty_of_primes](#)âĈĒ

Beweis. Consider the following curious topology on the set $\mathbb{Z}$ of integers. For $a, b \in \mathbb{Z}, b > 0$, we set

$$N_{a,b} = \{a + nb : n \in \mathbb{Z}\}.$$

Each set $N_{a,b}$ is a two-way infinite arithmetic progression. Now call a set $O \subseteq \mathbb{Z}$ open if either O is empty, or if to every $a \in O$ there exists some $b > 0$ with $N_{a,b} \subseteq O$. Clearly, the union of open sets is open again. If O_1, O_2 are open, and $a \in O_1 \cap O_2$ with $N_{a,b_1} \subseteq O_1$ and $N_{a,b_2} \subseteq O_2$, then $a \in N_{a,b_1 b_2} \subseteq O_1 \cap O_2$. So we conclude that any finite intersection of open sets is again open. Therefore, this family of open sets induces a bona fide topology on $\mathbb{Z}$.

Let us note two facts:

- (A) Any nonempty open set is infinite.
- (B) Any set $N_{a,b}$ is closed as well.

Indeed, the first fact follows from the definition. For the second, we observe

$$N_{a,b} = \mathbb{Z} \setminus \bigcup_{i=1}^{b-1} N_{a+i,b},$$

which proves that $N_{a,b}$ is the complement of an open set and hence closed.

So far, the primes have not yet entered the picture — but here they come. Since any number $n \neq 1, -1$ has a prime divisor p , and hence is contained in $N_{0,p}$, we conclude

$$\mathbb{Z} \setminus \{1, -1\} = \bigcup_{p \in \mathbb{P}} N_{0,p}.$$

Now if $\mathbb{P}$ were finite, then $\bigcup_{p \in \mathbb{P}} N_{0,p}$ would be a finite union of closed sets (by (B)), and hence closed. Consequently, $\{1, -1\}$ would be an open set, in violation of (A). $\square$

Theorem 1.6 (Sixth Proof). *The series $\sum_{p \in \mathbb{P}} \frac{1}{p}$ diverges.*

Beweis. Our final proof goes a considerable step further and demonstrates not only that there are infinitely many primes, but also that the series $\sum_{p \in \mathbb{P}} \frac{1}{p}$ diverges. The first proof of this important result was given by Euler (and is interesting in its own right), but our proof, devised by Erdős, is of compelling beauty.

Let $p_1, p_2, p_3, \dots$ be the sequence of primes in increasing order, and assume that $\sum_{p \in \mathbb{P}} \frac{1}{p}$ converges. Then there must be a natural number k such that $\sum_{i \geq k+1} \frac{1}{p_i} < \frac{1}{2}$. Let us call $p_1, \dots, p_k$ the small primes, and $p_{k+1}, p_{k+2}, \dots$ the big primes. For an arbitrary natural number N , we therefore find

$$\sum_{i \geq k+1} \frac{N}{p_i} < \frac{N}{2}. \quad (1)$$

Let N_b be the number of positive integers $n \leq N$ which are divisible by at least one big prime, and N_s the number of positive integers $n \leq N$ which have only small prime divisors. We are going to show that for a suitable N

$$N_b + N_s < N,$$

which will be our desired contradiction, since by definition $N_b + N_s$ would have to be equal to N .

To estimate N_b , note that $\left\lfloor \frac{N}{p_i} \right\rfloor$ counts the positive integers $n \leq N$ which are multiples of p_i . Hence by (1) we obtain

$$N_b \leq \sum_{i \geq k+1} \left\lfloor \frac{N}{p_i} \right\rfloor < \frac{N}{2}. \quad (2)$$

Let us now look at N_s . We write every $n \leq N$ which has only small prime divisors in the form $n = a_n b_n^2$, where a_n is the square-free part. Every a_n is thus a product of *different* small primes, and we conclude that there are precisely 2^k different square-free parts. Furthermore, as $b_n^2 \leq n \leq N$, we find that there are at most $\sqrt{N}$ different square parts, and so

$$N_s \leq 2^k \sqrt{N}.$$

Since (2) holds for any N , it remains to find a number N with $2^k \sqrt{N} < \frac{N}{2}$, or $2^{k+1} < \sqrt{N}$, and for this $N = 2^{2k+2}$ will do. $\square$

1.1 Appendix: Infinitely many more proofs

Theorem 1.7. *If the sequence $S = (s_1, s_2, s_3, \dots)$ is almost injective and of subexponential growth, then the set $\mathbb{P}_S$ of primes that divide some member of S is infinite.*

Beweis. We may assume that $f(n)$ is monotonely increasing. Otherwise, replace $f(n)$ by $F(n) = \max_{i \leq n} f(i)$; you can easily check that with this $F(n)$ the sequence S again satisfies the subexponential growth condition.

Let us suppose for a contradiction that $\mathbb{P}_S = \{p_1, \dots, p_k\}$ is finite. For $n \in \mathbb{N}$, let

$$s_n = \varepsilon_n p_1^{\alpha_1} \cdots p_k^{\alpha_k}, \quad \text{with } \varepsilon_n \in \{1, 0, -1\}, \alpha_i \geq 0,$$

where the $\alpha_i = \alpha_i(n)$ depend on n . (For $s_n = 0$ we can put $\alpha_i = 0$ for all i .) Then

$$2^{\alpha_1 + \dots + \alpha_k} \leq |s_n| \leq 2^{f(n)} \quad \text{for } s_n \neq 0,$$

and thus by taking the binary logarithm

$$0 \leq \alpha_i \leq \alpha_1 + \dots + \alpha_k \leq 2^{f(n)} \quad \text{for } 1 \leq i \leq k.$$

Hence there are not more than $2^{f(n)} + 1$ different possible values for each $\alpha_i = \alpha_i(n)$. Since f is monotone, this gives a first estimate

$$\#\{\text{distinct } |s_n| \neq 0 \text{ for } n \leq N\} \leq (2^{f(N)} + 1)^k \leq 2^{(f(N)+1)k}.$$

On the other hand, since S is almost injective only c terms in the sequence can be equal to 0, and each nonzero absolute value can occur at most $2c$ times, so we get the lower estimate

$$\#\{\text{distinct } |s_n| \neq 0 \text{ for } n \leq N\} \geq \frac{N - c}{2c}.$$

Altogether, this gives

$$\frac{N - c}{2c} \leq 2^{k(f(N)+1)}.$$

Taking again the logarithm with base 2 on both sides, we obtain

$$\log_2(N - c) - \log_2(2c) \leq k(f(N) + 1) \quad \text{for all } N.$$

This, however, is plainly false for large N , as k and c are constants, so $\frac{\log_2(N-c)}{\log_2 N}$ goes to 1 for $N \rightarrow \infty$, while $\frac{f(N)}{\log_2 N}$ goes to 0. □

Theorem 1.8 (Infinity of primes). *There are infinitely many primes. (Six + infinitely many proofs)*

Beweis. See theorems in this chapter. □

Uses:
thm:euclids_proof,
thm:second_proof,
thm:third_proof,
thm:fourth_proof,
thm:fifth_proof,
thm:sixth_proof,
thm:infy_proof